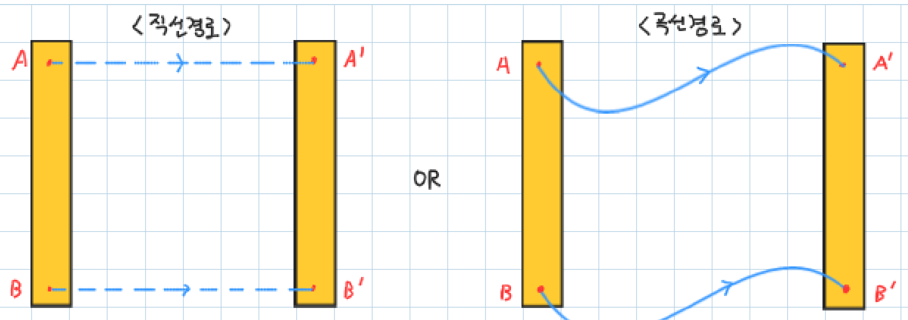
Chapter 2. 동적시스템의 모델링

1. 기계적 시스템 요소의 모델링(병진운동)

— 병진운동(Translational Motion)

• 직선 or 곡선을 따라 일어나는 운동

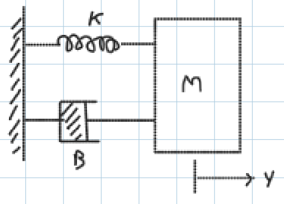


— Neton's 2nd Law

• $\Sigma \vec{F} = m\vec{a}$

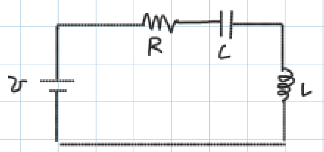
(+) ┌ 소문자 : 시간에 대해 변함
　　└ 대문자 : 시간에 대해 일정

— 기계적 시스템의 기본 구성요소 vs 전기시스템 기본 구성요소(R-L-C)



질량이 큰 물체는 움직이는 데 큰 힘이 듦

┌ mass
├ spring　　물체의 움직임을 방해하는 요소
└ damper



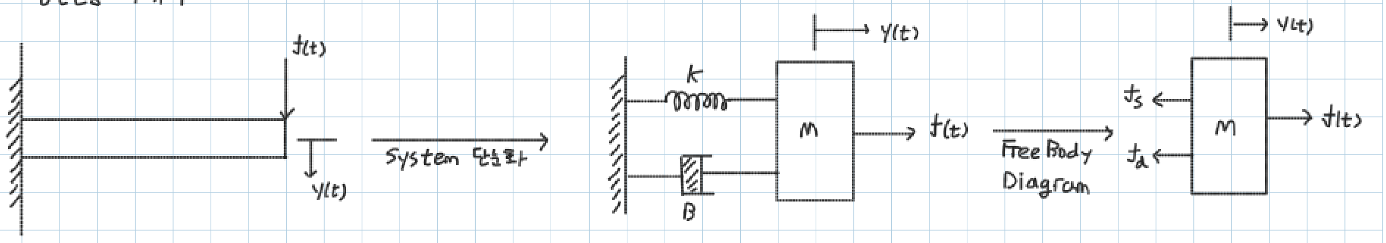
┌ Register
├ inductor
└ Capacitor

Input : Force　Output : position or velocity

Input : Voltage　Output : Current or 전하량

┌ f_s (스프링에 의한 힘) $= Ky(t)$, K : spring 상수
└ f_d (damper에 의한 힘) $= B\frac{dy(t)}{dt}$, B : 점성마찰 계수

— 병진운동 예제 1



$\Sigma F = ma$: $f(t) - f_s - f_d = Ma \rightarrow M\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Ky = f(t) \rightarrow \ddot{y} + \frac{B}{M}\dot{y} + \frac{K}{M}y = \frac{1}{M}f(t)$

$\rightarrow \ddot{y} + \frac{B}{M}\dot{y} + \frac{K}{M}y = \frac{K}{M}\Big(\underbrace{\frac{f(t)}{K}}_{r(t)}\Big) \rightarrow \boxed{\ddot{y} + \frac{B}{M}\dot{y} + \frac{K}{M}y = \frac{K}{M}r(t)}$

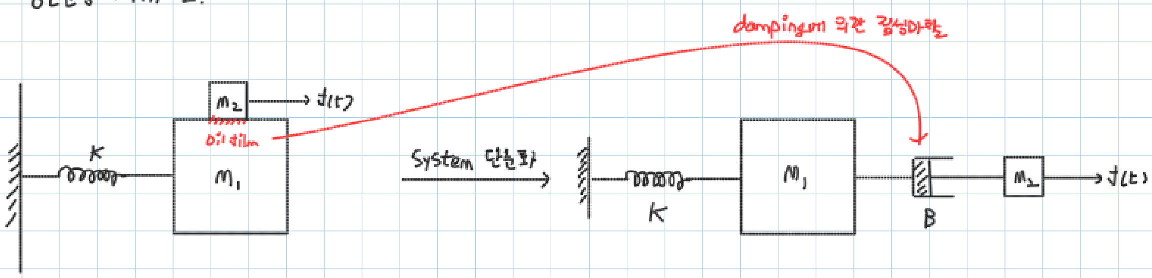
$$\boxed{\begin{array}{l}\ddot{y} + 2\zeta\omega_n\dot{y} + \omega_n^2 y = \omega_n^2 r(t) \\ \omega_n : \text{natural frequency (고유주파수)} , \ \zeta : \text{damping ratio (감쇠비)}\end{array}}$$

prototype second order System (표준형 2차 시스템)

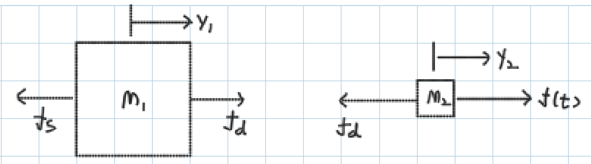
• 기계, 전기, 유체, 열 시스템 모두 prototype second order System 형태로 표현됨

∴ prototy second order system에 대한 Control 하면 기계, 전기, 유체, 열 시스템에 대해 적용

— 병진운동 예제 2.



Free Body Diagram



$f_s = Ky_1 \quad f_d = B(\dot{y}_2 - \dot{y}_1)$

(i) M_1

$\xrightarrow{+} \sum F = ma$: $f_d - f_s = m_1\ddot{y}_1 \longrightarrow B(\dot{y}_2 - \dot{y}_1) - Ky_1 = m_1\ddot{y}_1$

$m_1\ddot{y}_1 + B(\dot{y}_1 - \dot{y}_2) + Ky_1 = 0$

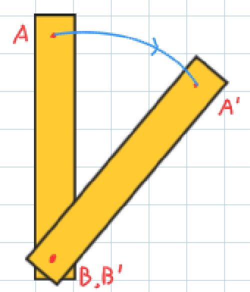
(ii) m_2

• $\xrightarrow{+} \sum F = ma$: $f(t) - f_d = m_2\ddot{y}_2 \longrightarrow f(t) - B(\dot{y}_2 - \dot{y}_1) = m_2\ddot{y}_2$

$m_2\ddot{y}_2 + B(\dot{y}_2 - \dot{y}_1) = f(t)$

2. 기계적 시스템 요소의 모델링(회전 운동)

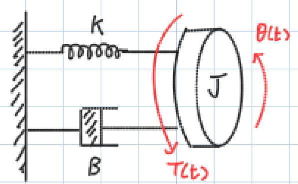
— 회전운동(rotational motion)



— Newton's 2nd Law (rotation)

• $\sum T = J\alpha$, J : moment of inertia (관성모멘트) α : 각가속도

— 회전운동의 기본 구성요소

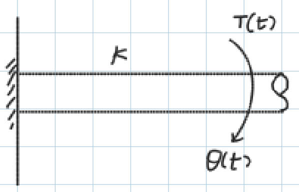


관성모멘트 크면 회전시키기 어려움 ($J = \int r^2 dm$)

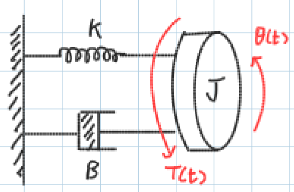
- 관성모멘트
- 비틀림 스프링 : $T_s = K\theta(t)$
- 댐퍼 : $T_d = B\dot{\theta}(t)$

회전운동 방해하는 요소

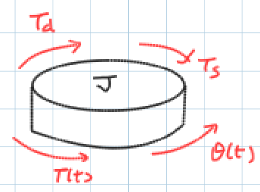
— 회전운동 예제 1



System 단순화



Free Body Diagram



• $\sum T = J\alpha$: $T(t) - \underbrace{\cancel{T_d}}_{B\dot{\theta}} - \underbrace{\cancel{T_s}}_{K\theta} = J\ddot{\theta} \longrightarrow J\ddot{\theta} + B\dot{\theta} + K\theta = T(t) \longrightarrow \ddot{\theta} + \frac{B}{J}\dot{\theta} + \frac{K}{J}\theta = \frac{1}{J}T(t)$

$\ddot{\theta} + \frac{B}{J}\dot{\theta} + \frac{K}{J}\theta = \frac{K}{J}\left(\overbrace{\frac{T(t)}{K}}^{r(t)}\right)$

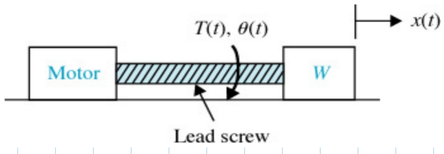
$\updownarrow$

$\boxed{\ddot{\theta} + 2\zeta\omega_n\dot{\theta} + \omega_n^2\theta = \omega_n^2 r(t)}$

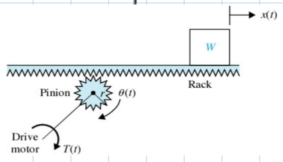
prototype second order system

3. 병진운동과 회전운동 사이의 변환

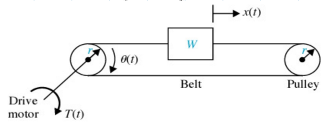
① lead screw



② rack and pinion



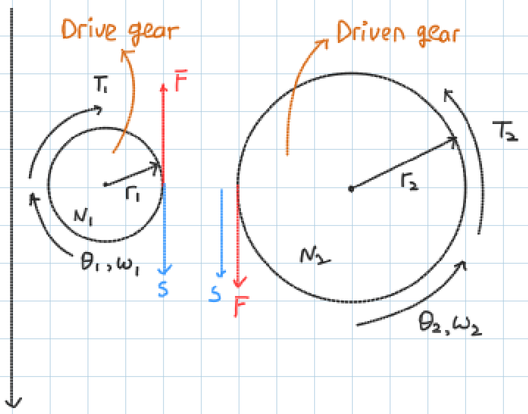
③ belt and pulley



4. 기어 열 (Gear)

— 기어(Gear)

• 에너지를 전달하는 장치로 힘, 토크, 속력, 변위가 변함



☆

$$\boxed{\frac{r_2}{r_1} = \frac{N_2}{N_1} = \frac{T_2}{T_1} = \frac{\theta_1}{\theta_2} = \frac{\omega_1}{\omega_2}}$$

proof

• assumption : 기어가 등속도로 회전한다고 가정

$T_1 = r_1 F$, $T_2 = r_2 F$

$$\frac{T_1}{T_2} = \frac{r_1 F}{r_2 F} = \frac{r_1}{r_2}$$

$s = r_1\theta_1 = r_2\theta_2$

$$\frac{r_1}{r_2} = \frac{\theta_2}{\theta_1}$$

$$v = \frac{d}{dt}(r_1\theta_1) = \frac{d}{dt}(r_2\theta_2)$$

$$= r_1\omega_1 = r_2\omega_2$$

$$\frac{r_1}{r_2} = \frac{\omega_2}{\omega_1}$$

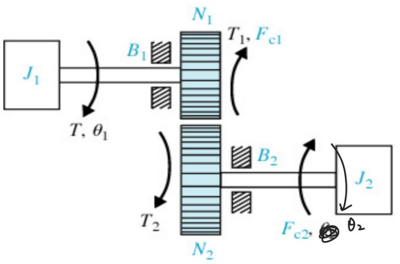
ex) 모터의 회전속도는 엄청 빠르지만 토크는 작음

↓ 기어를 통해 (1 : 100)

토크는 커지지만 회전속도는 감소

but, 원래 회전속도가 너무 빨라 1/100로 줄여도 OK.

5. 마찰과 관성을 가지는 기어(gear)



(i) FBD 1

• $\Sigma T = J\alpha$: $T - T_1 - B_1\dot{\theta}_1 - F_{c1}\dfrac{\omega_1}{|\omega_1|} = J_1\ddot{\theta}_1$

$$\underline{J_1\ddot{\theta}_1 + B_1\dot{\theta}_1 + F_{c1}\frac{\omega_1}{|\omega_1|} + T_1 = T} \,//$$

(ii) FBD 2

• $\Sigma T = J\alpha$: $T_2 - B_2\dot{\theta}_2 - F_{c2}\dfrac{\omega_2}{|\omega_2|} = J_2\ddot{\theta}_2$

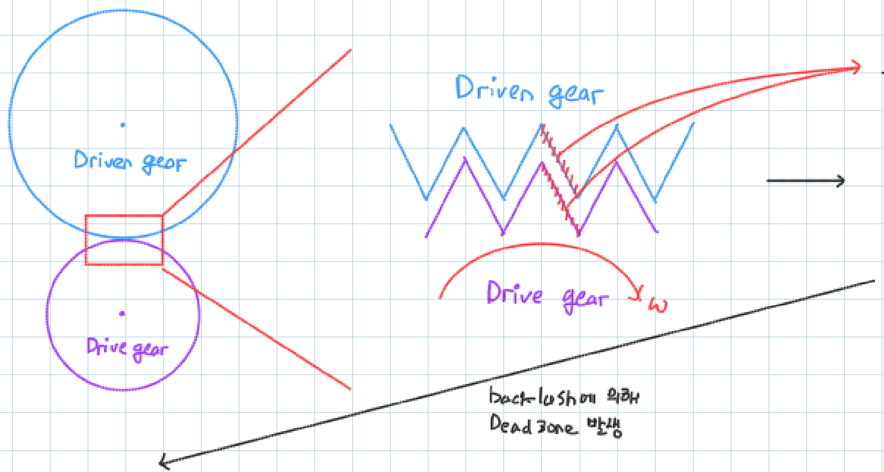
$$\underline{J_2\ddot{\theta}_2 + B_2\dot{\theta}_2 + F_{c2}\frac{\omega_2}{|\omega_2|} = T_2} \,//$$

(iii) $\underline{\dfrac{T_2}{T_1} = \dfrac{N_2}{N_1} = \dfrac{r_2}{r_1} = \dfrac{\theta_1}{\theta_2} = \dfrac{\omega_1}{\omega_2}} \,//$

6. Backlash 와 Dead Zone

— gear가 잘 돌아가기 위해서는?

- Drive gear와 Driven gear가 살짝 떨어져야 함 (∵ 너무 딱 맞으면 끼여서 안 돌아감)



Ex) 모터에 연결된 바퀴 살짝 돌릴 때 걸럭이는 느낌

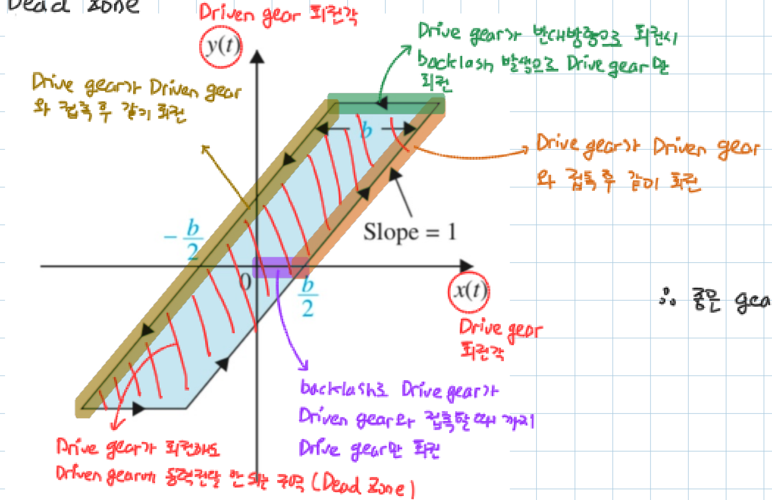
— backlash 발생

- Drive gear가 회전하지만 Driven gear와 살짝 떨어져 있어 Driven gear에 접촉할 때까지 Drive gear만 회전

  ∴ 동력전달 X

- backlash가 크면 안 좋음

— Dead Zone



∴ 좋은 gear : backlash가 없거나 Dead zone이 작은 gear

Dead zone = 0

ex) harmonic drive

7. 전기회로망에 관한 모델링

- KVL (Kirchhoff's Voltage Law), KCL (Kirchhoff's Current Law), 수동소자 (R, L, C) 법칙을 이용해 modeling

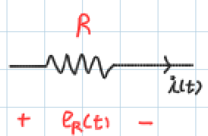
KVL

- 폐회로 에서 Σ 전압상승 = Σ 전압강하

KCL

- 특정 node에 대해 $\Sigma i_{in} = \Sigma i_{out}$

— Registors

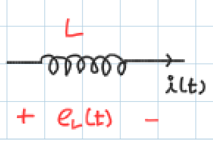


- $e_R(t) = i(t)R$

(*) e : electromotive force (= voltage)

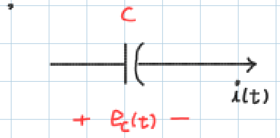
기전력 (전기를 발생시키는 힘)

— Inductors



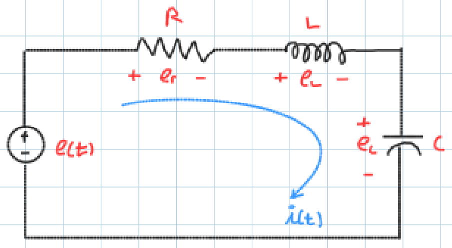
- $e_L(t) = L\dfrac{di(t)}{dt}$

— Capacitor



• $e_c(t) = \frac{1}{C}\int i(t)dt$

— 예제 2-2-1



• KVL

$$e(t) = e_r + e_L + e_c$$

$$= Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt$$

$$\frac{d}{dt}(e(t)) = \frac{d}{dt}\left[Ri(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)dt\right]$$

$$\dot{e}(t) = R\frac{d}{dt}i(t) + L\frac{d^2}{dt^2}i(t) + \frac{1}{C}i(t)$$

$$\rightarrow L\frac{d^2i(t)}{dt^2} + R\frac{di(t)}{dt} + \frac{1}{C}i(t) = \dot{e}(t)$$

protstype 2nd order system

$$\frac{d^2i(t)}{dt^2} + \frac{R}{L}\frac{di(t)}{dt} + \frac{1}{LC}i(t) = \frac{1}{L}\dot{e}(t)$$

$$\ddot{i}(t) + 2\zeta\omega_n\dot{i}(t) + \omega_n^2 i(t) = \omega_n^2\dot{e}(t)$$

translational mechanical system 과 유사

8. 비선형시스템의 선형화 (Linearization)

• 대부분의 시스템은 비선형 시스템 ⟶ Taylor Series 를 이용해 관심있는 점 주변을 선형화.

비선형 시스템 전체 선형화 X

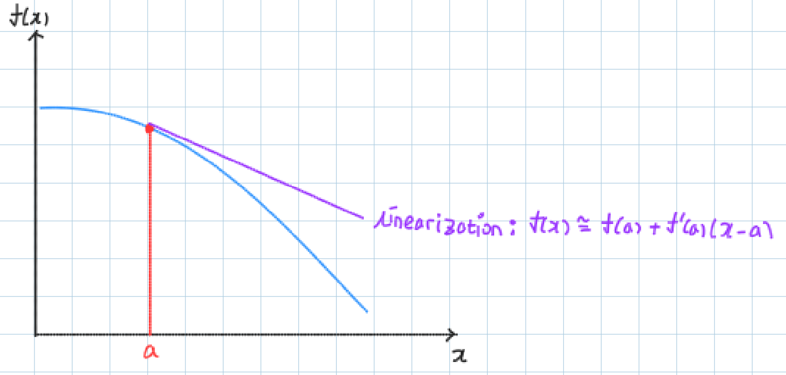
— Taylor Series

• 모든 함수는 taylor Series 로 표현가능

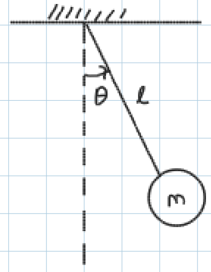
$$f(x) = \sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n = \lim_{n\to\infty}\left[f(a) + \frac{df(a)}{dt}(x-a) + \frac{1}{2}\frac{d^2f(a)}{dt^2}(x-a)^2 + \frac{1}{3!}\frac{d^3f(a)}{dt^3}(x-a)^3 + \cdots + \frac{1}{n!}\frac{d^nf(a)}{dt^n}(x-a)^n\right]$$

— 비선형 함수($f(x)$)의 관심있는 점 a의 주변을 taylor Series의 첫번째, 두 번째항만 이용해 근사화 (선형화)

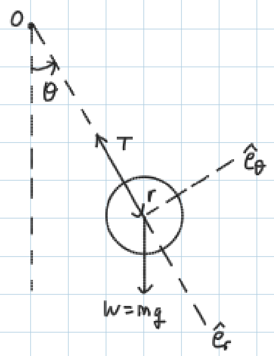
• $f(x) \doteqdot f(a) + \frac{df(a)}{dt}(x-a)$



9. 비선형시스템의 선형화(Linearization) — 예제



→ modeling?



- $\vec{r} = r\hat{e}_r$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(r\hat{e}_r) = \dot{r}\hat{e}_r + r\dot{\hat{e}}_r$$

$$= \dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta$$

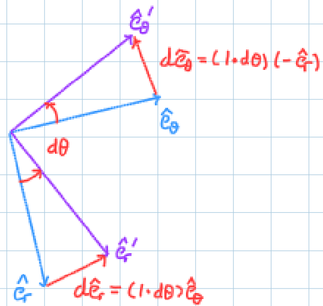
$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\dot{r}\hat{e}_r + r\dot{\theta}\hat{e}_\theta)$$

$$= \ddot{r}\hat{e}_r + \dot{r}\underbrace{\cancel{\dot{\hat{e}}_r}}_{\dot{\theta}\hat{e}_\theta} + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta + r\dot{\theta}\underbrace{\cancel{\dot{\hat{e}}_\theta}}_{-\dot{\theta}\hat{e}_r}$$

$$= \ddot{r}\hat{e}_r + \dot{r}\dot{\theta}\hat{e}_\theta + \dot{r}\dot{\theta}\hat{e}_\theta + r\ddot{\theta}\hat{e}_\theta - r\dot{\theta}^2\hat{e}_r$$

$$= (\ddot{r} - r\dot{\theta}^2)\hat{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{e}_\theta$$

$$\begin{cases} a_r = \ddot{r} - r\dot{\theta}^2 \\ a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta} \end{cases} \xrightarrow[r=\text{Const}]{} \begin{matrix} a_r = -r\dot{\theta}^2 \\ a_\theta = r\ddot{\theta} \end{matrix}$$



- $\dot{\hat{e}}_r = \frac{d\hat{e}_r}{dt} = \frac{(1\cdot d\theta)\hat{e}_\theta}{dt} = \dot{\theta}\hat{e}_\theta$
- $\dot{\hat{e}}_\theta = \frac{d\hat{e}_\theta}{dt} = \frac{(1\cdot d\theta)(-\hat{e}_r)}{dt} = -\dot{\theta}\hat{e}_r$

(i)

- $\Sigma F_r = ma_r$: $-T + mg\cos\theta = m(-r\dot{\theta}^2)$

$$= -m\underbrace{\cancel{r}}_{\ell}\dot{\theta}^2$$

$$= -m\ell\dot{\theta}^2$$

(ii)

- $\Sigma F_\theta = ma_\theta$: $-\cancel{m}g\sin\theta = \cancel{m}\underbrace{\cancel{r}}_{\ell}\ddot{\theta}$

$$\ell\ddot{\theta} = -g\sin\theta$$

$$\ell\ddot{\theta} + g\sin\theta = 0$$

$$\underline{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0}$$

but 비선형

선형화(linearization)

"taylor Series"

- $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n = \lim_{n\to\infty}\left[f(a) + \frac{df(a)}{dt}(x-a) + \frac{1}{2}\frac{d^2f(a)}{dt^2}(x-a)^2 + \cdots + \frac{1}{n!}\frac{d^nf(a)}{dt}(x-a)^n\right]$

$$f(x) \cong f(a) + \frac{df(a)}{dt}(x-a)$$

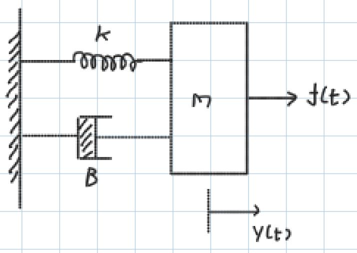
$$\sin x \cong \sin a + \cos a(x-a) \xrightarrow[\substack{a=0 \\ (\text{assumption})}]{} \sin x \cong x$$

$$\underline{\ddot{\theta} + \frac{g}{\ell}\sin\theta = 0} \xrightarrow[\text{linearization}]{} \underline{\ddot{\theta} + \frac{g}{\ell}\theta = 0 \quad (\theta\text{가 0주변에 있을 때})}$$

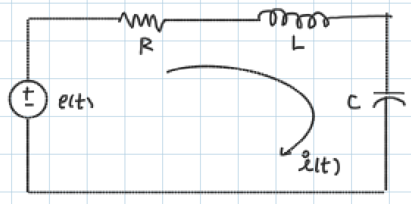
$$\boxed{\begin{aligned} \therefore\ & \ddot{\theta} + \frac{g}{\ell}\sin\theta = 0 \\ & \Big\downarrow \text{linearization} \\ & \ddot{\theta} + \frac{g}{\ell}\theta = 0 \quad (\theta\text{가 0주변에 있을 때}) \end{aligned}}$$

10. 기계시스템과 전기회로망의 유사성

① mechanical system (spring-mass-damper)



② electronic system (RLC network)



— modeling

- $m\frac{d^2y}{dt^2} + B\frac{dy}{dt} + ky = f(t)$

— modeling

- $L\frac{d^2i(t)}{dt^2} + R\frac{di(t)}{dt} + \frac{1}{C}i(t) = \frac{d}{dt}e(t)$

$\longleftrightarrow$ 유사

prototy. 2nd order ODE

- $\ddot{x} + 2\zeta\omega_n\dot{x} + \omega_n^2 x = \omega_n^2 u(t)$